Program: **Diploma in Communication and Computer Networking**	
Course Code: **6322C**	Course Title: **Data Communication**
Semester: **6**	Credits: **4**
Course Category: **Programme core course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- Understand the fundamentals of data communication
- Understand data conversions and error handling in data.
- Understand bandwidth utilization and switching.
- Understand about data link control and protocols.

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamentals of data communication.	14	Understanding
CO2	Understand data conversions and error handling in data	14	Applying
CO3	Understand bandwidth utilization and switching	14	Understanding
CO4	Understand about data link control and protocols	16	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2				
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the fundamentals of data communication**		
M1.01	Data and Signals – Analog and Digital – Data transmission – Basics – Transmission impairments – Data rate limits – performance	5	Understanding
M1.02	Digital transmission – Analog transmission –asynchronous transmission – synchronous transmission	4	Understanding
M1.03	signal propagation delay – transmission media - guided media – unguided media	5	Understanding
Contents: Data and Signals – Analog and Digital – Data transmission – Basics – Transmission impairments – Data rate limits – performance – Digital transmission – Analog transmission – asynchronous transmission – synchronous transmission – signal propagation delay – transmission media - guided media – unguided media			
CO2	**Understand data conversions and error handling in data.**		
M2.01	Digital to analog conversion – analog to digital conversion	3	Understanding
M2.02	transmission modes – error detection and correction	5	Understanding

M2.03	introduction – block coding – cyclic codes – checksum – data compression.	6	Understanding
	Series Test –I	1	
Contents: Digital to analog conversion – analog to digital conversion – transmission modes – error detection and correction – introduction – block coding – cyclic codes – checksum – data compression.			
CO3	**Understand bandwidth utilization and switching**		
M3.01	Bandwidth utilization – channel capacity	3	Understanding
M3.02	multiplexing – spread spectrum	4	Applying
M3.03	switching – circuit switched networks – datagram networks – virtual circuit networks – structure of a switch	7	Understanding
Contents: Bandwidth utilization – channel capacity – multiplexing – spread spectrum – switching – circuit switched networks – datagram networks – virtual circuit networks – structure of a switch			
CO4	**Understand about data link control and protocols**		
M 4.01	Data link control – framing – flow control – error control – protocol basics	8	Understanding
M 4.02	character oriented protocols– bit-oriented protocols – noiseless channels – noisy channels – HDLC – point to point protocol	8	Understanding
	Series Test – II	1	
Data link control – framing – flow control – error control – protocol basics – character-oriented protocols– bit oriented protocols – noiseless channels – noisy channels – HDLC – point to point protocol			

Text / Reference

T/R	Book Title/Author
T1	Behrouz A. Forouzan, Data Communications and Networking 5E, 5th Edition,

	Tata McGraw-Hill, 2013.
R1	Alberto Leon-Garcia and Indra Widjaja: Communication Networks - Fundamental Concepts and Key architectures, 2nd Edition Tata McGraw-Hill, 2004.
R2	2. William Stallings: Data and Computer Communication, 8th Edition, Pearson Education, 2007.
R3	3. Larry L. Peterson and Bruce S. Davie: Computer Networks – A Systems Approach, 4th Edition, Elsevier, 2007.
R4	4. Nader F. Mir: Computer and Communication Networks, Pearson Education, 2007.